

Robust Offline Safe RL under Dynamic Safety Budgets

Toward Soft & Calibrated OOD Regularization

RL Mid-term Presentation

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Background: Offline Safe RL

Traditional RL learns by repeatedly interacting with the environment.

In safety-critical domains, online exploration can be too expensive or risky, so we want to learn from an offline dataset.

Robotics

Avoid collisions, falls, and unsafe motions

Autonomous Driving

Unsafe exploration cannot be used to collect data

Healthcare

Policies must satisfy strict risk budgets

Problem Setup

Offline Safe RL optimizes reward while constraining cumulative safety cost:

$$\max_{\pi} \mathbb{E}_{\pi}[R(\tau)] \quad \text{s.t.} \quad \mathbb{E}_{\pi}[C(\tau)] \leq \epsilon$$

It must solve three problems at the same time:

1. OOD state-action pairs caused by offline data
2. A policy that satisfies the cost constraint
3. A safety budget that may change during deployment

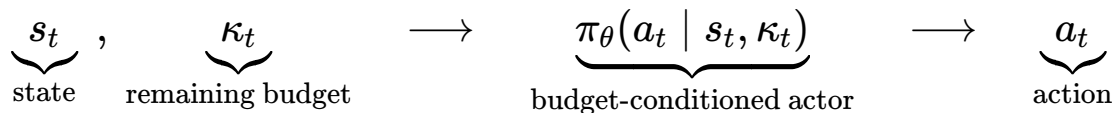
Goal: learn a high-reward and safe policy that can adapt to different remaining budgets.

CCAC: Existing Backbone

CCAC addresses this setting with a budget-conditioned actor-critic framework.

$$\pi_{\theta}(a_t \mid s_t, \kappa_t), \quad \kappa_{t+1} = \kappa_t - c_t$$

The remaining budget κ_t becomes part of the policy input, so the same policy can behave differently under different safety budgets.



Take-away: CCAC turns a fixed-threshold safe RL problem into a dynamic-budget policy learning problem.

CCAC: How Safety Is Enforced

The key mechanism is to identify generated OOD / unsafe state-action pairs and penalize them through the cost critic.

Module	Role	Notation
Offline data	Build budget-annotated transitions	$D = \{(s_t, a_t, r_t, c_t, s_{t+1}, \kappa_t)\}$
Constraint-conditioned CVAE	Generate state-action pairs under a budget	$(\hat{s}, \hat{a}) \sim p_\phi(s, a \mid z, \kappa)$
OOD classifier	Decide whether (s, a, κ) is OOD / unsafe	$h_\psi(s, a \mid \kappa) > 0.5$
Cost critic	Overestimate cost for unsafe samples	$Q_c(s, a \mid \kappa)$
Actor	Maximize reward while avoiding high cost	$\pi_\theta(a \mid s, \kappa)$

$$p_\phi(s, a \mid z, \kappa) \rightarrow h_\psi(s, a \mid \kappa) \rightarrow Q_c(s, a \mid \kappa) \rightarrow \pi_\theta(a \mid s, \kappa)$$

Our Observation: A Brittle OOD Gate

CCAC relies heavily on the OOD classifier. If classifier decisions are inaccurate, cost overestimation can become unstable.

The original hard OOD rule is:

$$\mathbb{I}_{\text{OOD}}(s, a, \kappa) = \mathbb{I}[h_{\psi}(s, a \mid \kappa) > 0.5]$$

This can cause two issues:

1. Classifier scores may not be calibrated; 0.49 and 0.51 are treated completely differently
2. Under imbalanced data, the classifier may miss truly dangerous OOD samples

Question: can we make OOD regularization softer and more calibrated?

Our Small Improvement

Soft & Calibrated OOD Regularization

Instead of hard filtering, use a soft OOD weight:

$$w_{\text{OOD}}(s, a, \kappa) = \sigma(\alpha(h_{\psi}(s, a \mid \kappa) - \tau))$$

Then use w_{OOD} to weight the cost overestimation penalty, rather than making a binary decision.

We also improve classifier training with:

- weighted BCE or focal loss for class imbalance
- calibration analysis on classifier scores
- false negative rate analysis for unsafe OOD samples

Preliminary Validation

Small-scale experiments on `OfflineBallRun-v0` give a first feasibility signal.

Test	Result	Interpretation
Classifier-only	OOD FNR: 10.4% $\rightarrow$ 3.6% with focal loss	Fewer dangerous OOD samples are missed
Cost critic separation	<code>focal_soft</code> matched gap: 8.71	OOD samples receive higher cost values than matched IND samples
Full policy, 50k steps	zero realized cost for budgets 1/2/5/10	Variant can be integrated into full policy training

Current limitation: this is one task and one seed, so it supports feasibility rather than a final performance claim.

Feasibility Evidence

Compared with the 50k original CCAC baseline, `focal_soft` keeps zero realized cost and improves reward.

Target cost	Original CCAC reward / cost	<code>focal_soft</code> reward / cost	Reward gain
1	328/0	418/0	+27.5%
2	345/0	401/0	+16.2%
5	354/0	421/0	+19.1%
10	363/0	451/0	+24.4%

Take-away: the modification is small, trainable, and shows a positive reward-cost signal in a preliminary full-policy run.

Next Steps for Final Presentation

The remaining work is to test whether this preliminary signal is robust.

Direction	Plan
More runs	repeat <code>focal_soft</code> on more seeds and at least one additional task
Calibration	report ECE and reliability curves; avoid claiming calibration improvement too early
Ablations	hard vs. soft OOD, BCE vs. focal loss, fixed vs. varying budgets
Policy metrics	reward, realized cost, violation rate, OOD FNR, budget adaptation

For the mid-term, the result is enough to show feasibility; the final presentation will focus on robustness.

Related Work

Direction	Methods	What they address	Remaining gap
Offline RL	CQL, IQL, BCQ, BEAR	Distribution shift	No explicit safety constraints
Safe RL	Lagrangian, CPO, PID-Lagrangian	Constrained optimization	Often needs online interaction
Offline Safe RL	CPQ, COptiDICE, VOCE, FISOR	Reward-cost trade-off from offline data	Hard to adapt to changing budgets
Budget-conditioned OSRL	CCAC	Dynamic budget adaptation + OOD detection	Classifier threshold may be brittle

Our angle: make the OOD safety signal softer, calibrated, and easier to evaluate.

Q&A

Thank you!

References

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